The main thing is not to oversalt --- Solution

W21 (2021) — English translation

A1 **Measure the resistance of the limiting resistor R_1 with an ohmmeter (see equipment). The approximate value of 10 ohms is indicated below, but use the more accurate value you obtained in your calculations.** ??

Let's measure the resistance of the limiting resistor $R_1 = 10,1\ Ом$ with a multimeter in ohmmeter mode.

A2 **Experimentally confirm the validity of the proposed model. To do this, study the current-voltage characteristics of the solution at 5 different distances between the electrodes.** 7.50

We assemble a measuring circuit consisting of a battery connected in series, a control unit and a syringe with a salt solution. We connect a voltmeter to the electrodes of the syringe. We connect the second voltmeter in parallel with the reference resistance R_1 to measure the current in the circuit.

The entire set of processes on the cathode and anode is proposed to be represented in the form of a series circuit consisting of an external voltage applied to it U_0, resistance of the electrolyte volume R, contact EMF $U_к$ and contact resistance $r_к$ (Fig. 1). It is assumed that the value R is directly proportional to the resistivity of the electrolyte ρ, the distance between the electrodes L and inversely proportional to the cross-sectional area of the electrolytic bath S.

The theoretical current-voltage characteristic of the electrolytic bath within the framework of this model has the form \rho ur_k$.

We take the current-voltage characteristic of the solution. The tables show the results of measuring the current-voltage characteristics of the solution at 5 different volumes of liquid in the syringe V (and the corresponding distances L between the electrodes).

U_{10} , mV U, mV I, mA V, ml L, cm 290 1740 28.7 4 1.3 320 1860 31.7 450 2060 44.6 650 2460 64.4 770 2670 76.2 990 3080 98.0 1080 3420 106.9

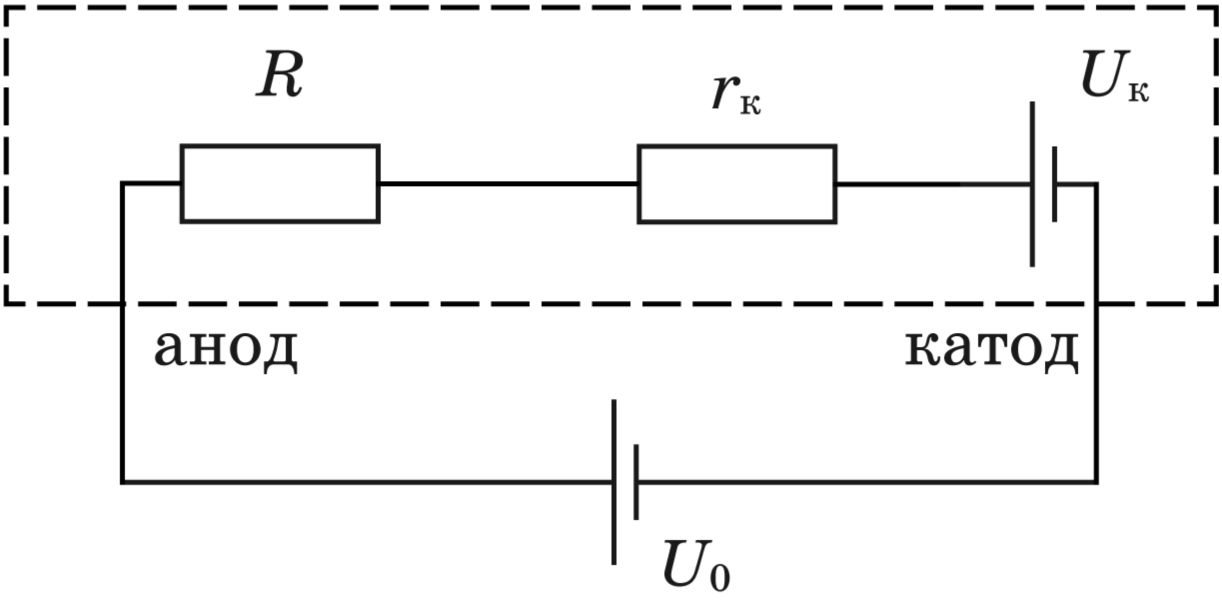
U_{10} , mV U, mV I, mA V, ml L, cm 870 3750 86.1 8 2.6 800 3570 79.2 740 3340 73.3 540 2920 53.5 392 2300 38.8 294 2040 29.1 269 1970 26.6

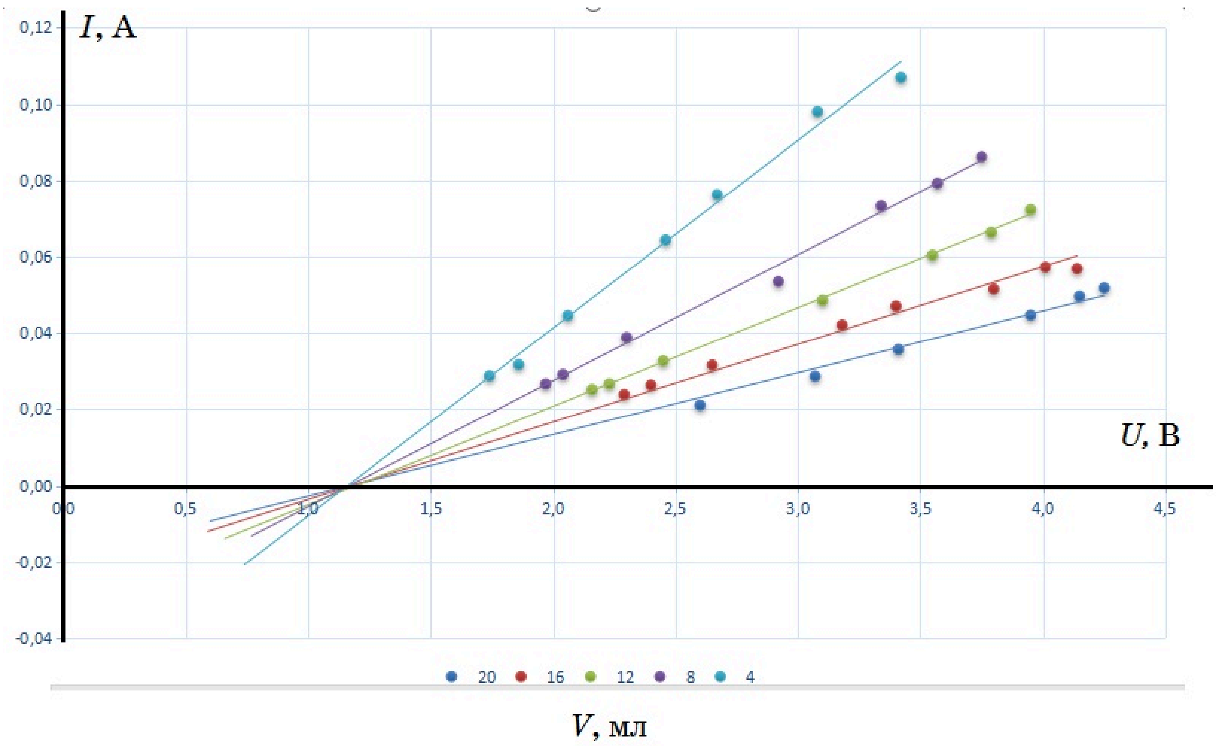
U_{10} , mV U, mV I, mA V, ml L, cm 254 2160 25.1 12 3.9 269 2230 26.6 330 2450 32.7 490 3100 48.5 610 3550 60.4 670 3790 66.3 730 3950 72.3

U_{10} , mV U, mV I, mA V, ml L, cm 574 4140 56.8 16 5.2 578 4010 57.2 520 3800 51.5 475 3400 47.0 425 3180 42.1 319 2650 31.6 266 2400 26.3

U_{10} , mV U, mV I, mA V, ml L, cm 213 2600 21.1 20 6.5 289 3070 28.6 360 3410 35.6 451 3950 44.7 501 4150 49.6 523 4250 51.8

We build the current-voltage characteristics for all experiments





A3 **Estimate the magnitude of the contact potential difference $U_к$.** **1.00**

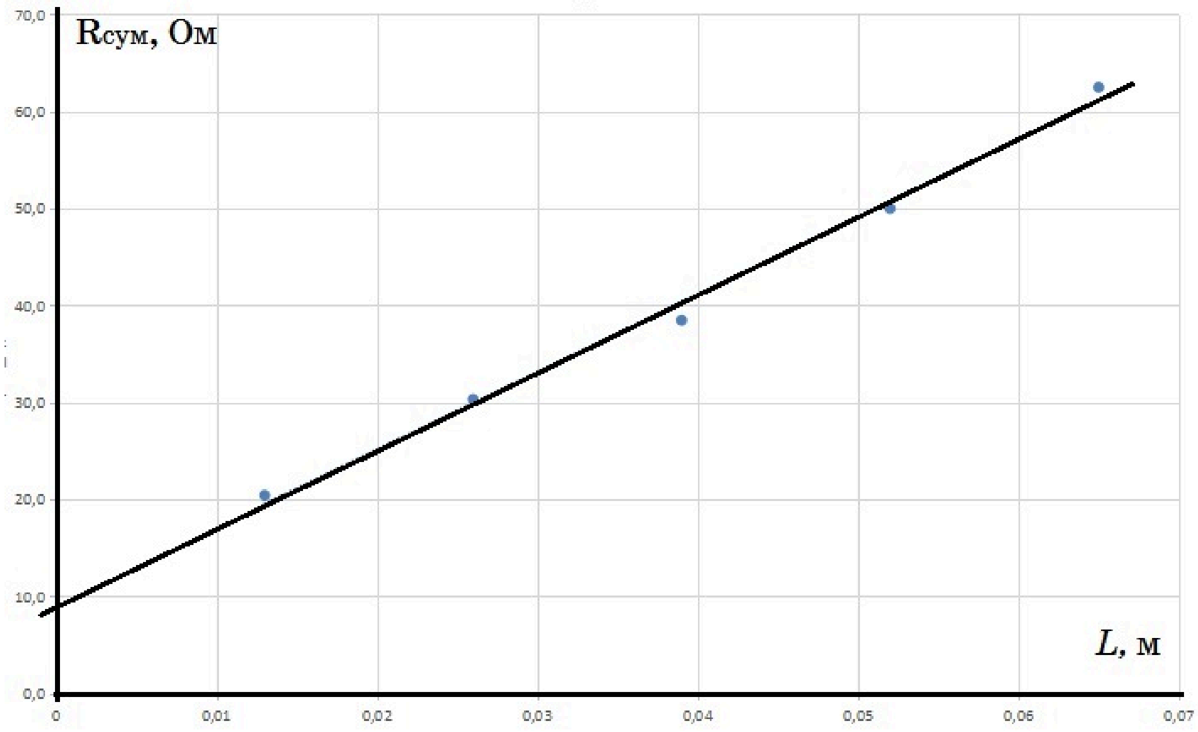
Using the graph, we determine: 1. From the point of intersection of the graphs and the voltage axis, the value of the contact potential difference $U_к = 1,2\ B$. 2. From the angular coefficient (see formula (2)) the resistance of the electrolyte volume together with the resistance of the contact areas $R_{сум}$ for 5 values L.

A4 **Determine the value of contact resistance $r_к$.** **2.50**

The results of processing experimental data for five values of L are presented in the table and figure.

No. L, m $1/R_{сум}$, 1/Ohm $R_{сум}$, Ohm 1 0.065 0.016 62.5 2 0.052 0.020 50.0 3 0.039 0.026 38.5 4 0.026 0.033 30.3 5 0.013 0.049 20.4

From the graph we find $r_к = R_{сум}(0) = 9,2\ Ом$.



A5 **Calculate the specific conductivity σ of a three percent solution of sodium chloride in water.** **1.00**

Cross-sectional area of the piston $S = 3,1 \cdot 10^{-4}\ м^2$ (can be found by measuring the diameter of the piston or the ratio of volume to length). Within the framework of the used model $\Delta R_{сум}/\Delta L = \rho/S$. From the slope of the straight line $R_{сум}(L)$ we find the resistivity of the salt solution used $\rho = 0,24\ Ом \cdot м$. That is, $\sigma = 1/\rho = 4,2\ (Ом \cdot м)^{-1}$.

A6 **Calculate the mobility μ of sodium and chlorine ions in this solution.** **3.00**

Specific conductivity of the electrolyte $\sigma = 1/\rho = q_1 n_1 \mu_1 + q_2 n_2 \mu_2$ where q_1 and q_2 are the modules of the electrical charges of positive and negative ions, respectively (in the case of sodium and chlorine ions $q_1 = q_2 = e$, where e is the elementary charge), n_1 and n_2 are their concentrations (in our case they are the same and equal to n), μ_1 and μ_2 are the mobilities of negative and positive ions in solution, which, according to the conditions of the problem, are also equal to each other $\mu_1 = \mu_2 = \mu$. Finally,

$$\mu = \frac{\sigma}{2en}$$

We calculate the concentration of sodium (or chlorine) ions in a three percent solution of table salt $NaCl$

$$n = N/V = \frac{m_{NaCl} N_a \rho_{раств}}{M_{NaCl} m_{раств}} = 0,03 \frac{N_a \rho_{раств}}{M_{NaCl}} = 3,2 \cdot 10^{26}\ м^{-3}$$

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We calculate the mobility:

$$\mu = \frac{\sigma}{(2en)} = 4,0 \cdot 10^{-8} \frac{м^2}{В \cdot с}$$

, which is in good agreement with the literature data.